# Oligopoly

EC 111 Principles of Microeconomics

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# Reading

- These slides are not a substitute for reading Mankiw, Chapter 18, Oligopoly.
- Required reading: Mankiw, *Principles of Microeconomics*, 11th edition, Chapter 18, "Oligopoly".

# What you must know from this chapter

1. **Conditions:** what makes a market an oligopoly.
2. **Cartels and collusion:** firms acting together as one monopolist.
3. **Why collusion breaks down:** each firm's best response, and the Nash equilibrium.
4. How the outcome under oligopoly compares with monopoly and perfect competition.

Part 1

# What Makes a Market an Oligopoly?

# Oligopoly

**Definition: Oligopoly**
A market with only a few sellers, each offering a product similar or identical to the others'.

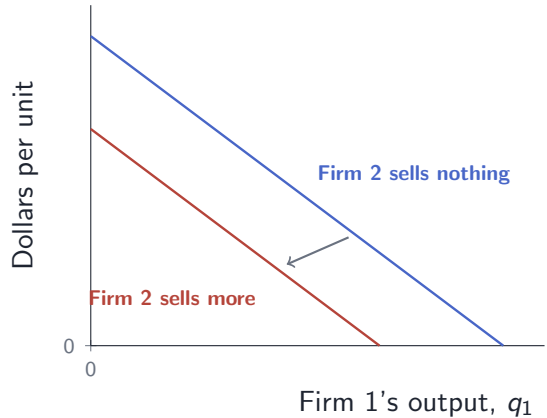
- **Few sellers:** a handful of firms supply most of the market.
- **Similar or identical products.**
- **Barriers to entry:** new firms find it hard to enter.
- **Interdependence:** each firm's profit depends on what its rivals do.

**Examples:** commercial aircraft (Boeing and Airbus); U.S. wireless phone service.

**Definition: Duopoly**
An oligopoly with exactly two sellers. This lecture studies a duopoly.

# Interdependence

- The two firms sell the same product at one market price.
- The price depends on the output of *both* firms.
- So the more firm 2 sells, the less demand is left for firm 1.
- Each firm must allow for what the other does.



# Four market structures

| | Perfect competition | Monopolistic competition | Oligopoly | Monopoly |
|---|---|---|---|---|
| Sellers | Many | Many | **A few** | One |
| Product | Identical | Differentiated | **Similar or identical** | No close substitutes |
| Entry | Free | Free | **Difficult** | Blocked |
| Rivals matter? | No | Little | **Yes** | No rivals |

**Finding**

What sets oligopoly apart is interdependence: each firm must allow for what its rivals do.

# Part 1 summary

- Oligopoly: a few sellers of similar or identical products.
- Entry is difficult.
- Interdependence: each firm's profit depends on its rival's output.
- The more one firm sells, the less demand is left for the other.
- Compared with perfect competition and monopoly: there, no rival's choice enters a firm's profit.

# Check: Part 1

Which of the following examples illustrates an oligopoly market?

A A farmers' market with many individuals selling sweet corn and tomatoes

B A city whose electrical service is provided by one electric co-operative

C A city with two firms that are licensed to sell school uniforms for the local schools

D A city with many independently owned hair styling salons

# Check: Part 1 – answer

Which of the following examples illustrates an oligopoly market?

A A farmers' market with many individuals selling sweet corn and tomatoes

B A city whose electrical service is provided by one electric co-operative

**C A city with two firms that are licensed to sell school uniforms for the local schools**

D A city with many independently owned hair styling salons

**Finding**

Two sellers of the same product, with entry blocked by licensing: a duopoly. A is perfect competition, B is monopoly, D is monopolistic competition.

Part 2

# Collusion: Acting as One Monopolist

# What collusion means

**Definition: Collusion**

An agreement among the firms in a market on the output each produces and the price they charge.

**Definition: Cartel**

A group of firms acting together. A central body sets total output for all of them.

- The cartel chooses output to maximize the firms' *joint* profit.
- So the firms act as one monopolist.

# Why the cartel can do better

- When firm 1 sells more, the market price falls.
- The lower price applies to firm 2's units too. That is a cost firm 1 imposes on firm 2.
- Acting alone, firm 1 ignores that cost: it is firm 2's loss, not firm 1's.
- The cartel counts it, because it maximizes the two firms' profit together.

**Finding**

Collusion makes each firm take account of the cost its output imposes on its rival. The firms then behave exactly like a monopolist.

# The set-up

Two firms, 1 and 2, sell an identical product.

- **Demand:** price equals average revenue,

$$P = a - bQ, \qquad Q = q_1 + q_2.$$

- **Cost:** each unit costs $c$, with no fixed cost, so $\text{TC} = c \times Q$.
- One more unit adds exactly $c$ to total cost, so $\text{MC} = c$.
- The cartel only has to choose total output $Q$. The split between the firms comes at the end.

# Recall from Chapter 16: MR for a straight-line demand curve

From the monopoly lecture, marginal revenue is $\text{MR} = P + Q \times \dfrac{\Delta P}{\Delta Q}$.

$$
\begin{aligned}
\text{MR} &= P + Q \times \frac{\Delta P}{\Delta Q} && \text{output effect plus price effect} \\
&= (a - bQ) + Q \times \frac{\Delta P}{\Delta Q} && \text{replace } P \text{ with } a - bQ \\
&= (a - bQ) + Q \times (-b) && \text{each extra unit lowers } P \text{ by } b, \text{ so } \tfrac{\Delta P}{\Delta Q} = -b \\
&= a - bQ - bQ && Q \times (-b) = -bQ \\
&= a - 2bQ && -bQ - bQ = -2bQ
\end{aligned}
$$

**Finding**

MR starts at the same point as demand, $a$, and is twice as steep: slope $-2b$ against $-b$.

# The cartel's output: MR = MC

The cartel maximizes joint profit, so it produces where MR = MC.

$$
\begin{aligned}
a - 2bQ &= c && \text{set MR} = \text{MC} \\
a - c &= 2bQ && \text{add } 2bQ \text{ to both sides; subtract } c \\
Q_C &= \frac{a-c}{2b} && \text{divide both sides by } 2b
\end{aligned}
$$

$Q_C$ is the collusive output: the monopoly output.

# The cartel's price

Read the price from the demand curve at $Q_C$.

$$
\begin{aligned}
P_C &= a - b \times \frac{a-c}{2b} && \text{put } Q_C \text{ into } P = a - bQ \\
&= a - \frac{a-c}{2} && \text{the } b\text{'s cancel} \\
&= \frac{2a-(a-c)}{2} = \frac{a+c}{2} && \text{write } a \text{ as } \tfrac{2a}{2}
\end{aligned}
$$

The price is halfway between $a$ and marginal cost $c$, so $P_C > c$.

# Splitting the output and the profit

The two firms are identical, so they split the cartel output equally.

$$q = \frac{Q_C}{2} = \frac{a-c}{4b} \qquad \text{each firm's output}$$

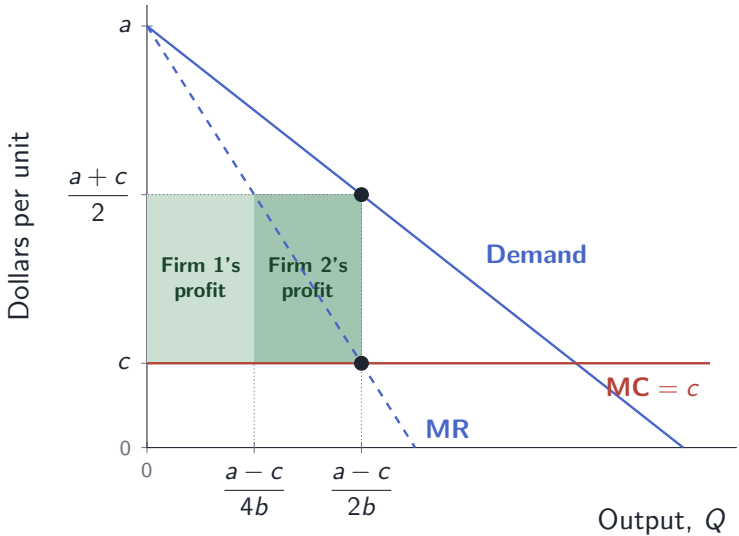
$$P_C - c = \frac{a+c}{2} - c = \frac{a-c}{2} \qquad \text{profit on each unit}$$

$$\pi = (P_C - c) \times q = \frac{a-c}{2} \times \frac{a-c}{4b} = \frac{(a-c)^2}{8b} \qquad \text{each firm's profit}$$

**Finding**

Each member produces $\frac{a-c}{4b}$ and earns $\frac{(a-c)^2}{8b}$: half the monopoly profit.

# The cartel on the graph



# Part 2 summary

- Collusion: the firms agree on output and price; a cartel sets total output.
- The cartel counts the cost each firm's output imposes on the other, so it acts as one monopolist.
- MR $= a - 2bQ = c$ gives $Q_C = \frac{a-c}{2b}$ and $P_C = \frac{a+c}{2}$.
- Each firm makes $\frac{a-c}{4b}$ and earns $\frac{(a-c)^2}{8b}$.
- Compared with perfect competition: less output, and a price above MC.

# Check: Part 2

If duopoly firms that are not colluding were able to successfully collude, then

A price and quantity would rise.

B price and quantity would fall.

C price would rise and quantity would fall.

D price would fall and quantity would rise.

# Check: Part 2 – answer

If duopoly firms that are not colluding were able to successfully collude, then

A price and quantity would rise.

B price and quantity would fall.

**C price would rise and quantity would fall.**

D price would fall and quantity would rise.

**Finding**

A cartel restricts output to the monopoly level, so quantity falls and price rises.

Part 3

# Oligopoly Without Collusion

# The key insight: price depends on both outputs

- Now there is no cartel. Each firm chooses its own output to maximize its own profit.
- The price depends on total output, $Q = q_1 + q_2$.
- So firm 1's profit depends on its own output *and* on firm 2's:

$$\pi_1 = (P - c) \times q_1 = \big(a - b(q_1 + q_2) - c\big) \times q_1.$$

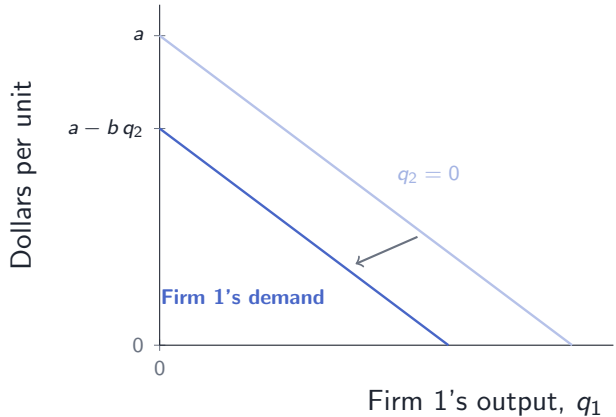
- Firm 1 takes firm 2's output $q_2$ as given and chooses $q_1$.

# The demand left for firm 1

Rearrange the price equation:

$$\begin{aligned} P &= a - b\,q_2 - b\,q_1 \\ &= (a - b\,q_2) - b\,q_1 \end{aligned}$$

- For a given $q_2$, this is a straight line in $q_1$.
- It starts at $a - b\,q_2$ and has slope $-b$.
- When $q_2$ rises, the line shifts down.



# Firm 1 maximizes profit: MR = MC

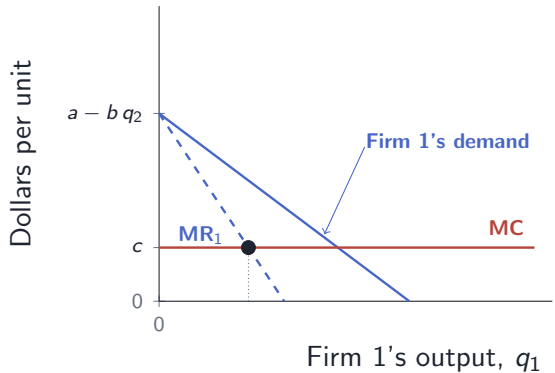
Firm 1's demand is a straight line starting at $a - b\,q_2$ with slope $-b$.

- By the Chapter 16 result, its MR starts at the same point and is twice as steep:

$$\text{MR}_1 = (a - b\,q_2) - 2b\,q_1.$$

- Firm 1 sets $\text{MR}_1 = \text{MC}$:

$$(a - b\,q_2) - 2b\,q_1 = c.$$



# Firm 1's best-response function

Solve $(a - b\,q_2) - 2b\,q_1 = c$ for $q_1$:

$$2b\,q_1 = a - c - b\,q_2 \qquad \text{add } 2b\,q_1 \text{ to both sides; subtract } c$$

$$q_1 = \frac{a-c}{2b} - \frac{q_2}{2} \qquad \text{divide both sides by } 2b$$

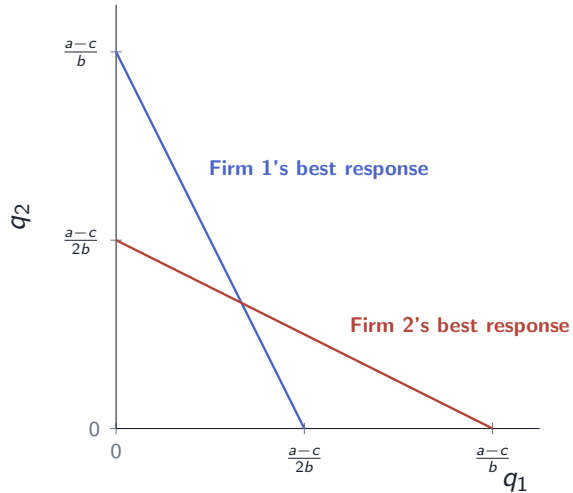
**Definition: Best-response function**

A firm's profit-maximizing output for each output its rival might choose.

- For each extra unit firm 2 produces, firm 1 produces half a unit less.
- **Firm 2**, by exactly the same steps: $q_2 = \frac{a-c}{2b} - \frac{q_1}{2}$.

# The best responses on one graph

- Axes: firm 1's output across, firm 2's up.
- Firm 1's line: if $q_2 = 0$, firm 1 makes the monopoly output $\frac{a-c}{2b}$.
- If firm 2 makes the competitive output $\frac{a-c}{b}$, firm 1 makes nothing.
- Firm 2's line is the mirror image.



# The Nash equilibrium

**Definition: Nash equilibrium**

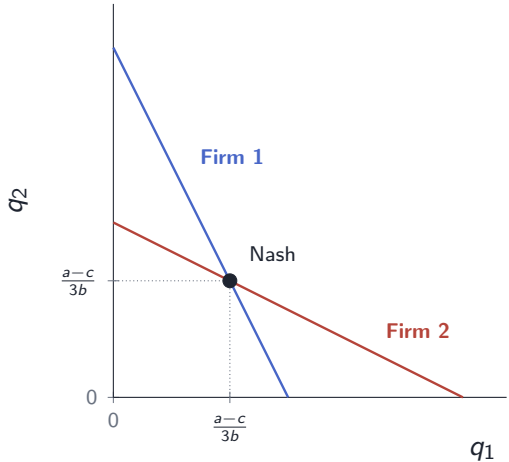
Each firm's output is its best response to the other's output.

It is where the two lines cross. The firms are identical, so $q_1 = q_2 = q$:

$$q = \frac{a-c}{2b} - \frac{q}{2}$$

$$\tfrac{3}{2}\, q = \frac{a-c}{2b}$$

$$q = \frac{a-c}{3b}$$



# Price and profit at the Nash equilibrium

$$Q = 2 \times \frac{a-c}{3b} = \frac{2(a-c)}{3b}$$ total output

$$P = a - b \times \frac{2(a-c)}{3b} = a - \frac{2(a-c)}{3} = \frac{a+2c}{3}$$ price from demand

$$P - c = \frac{a+2c}{3} - c = \frac{a-c}{3}$$ profit on each unit

$$\pi = \frac{a-c}{3} \times \frac{a-c}{3b} = \frac{(a-c)^2}{9b}$$ each firm's profit

**Finding**

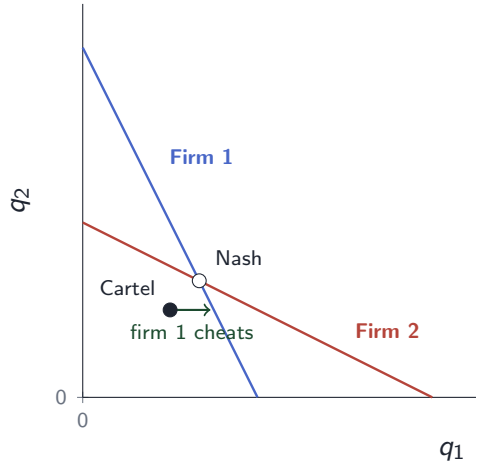
Each firm earns $\frac{(a-c)^2}{9b}$, less than the cartel's $\frac{(a-c)^2}{8b}$: one-ninth is less than one-eighth.

# Why the cartel breaks down

- Suppose firm 2 keeps to its cartel share, $\frac{a-c}{4b}$.
- Firm 1's best response:

$$q_1 = \frac{a-c}{2b} - \frac{1}{2} \times \frac{a-c}{4b} = \frac{3(a-c)}{8b}.$$

- That is more than its cartel share, $\frac{2(a-c)}{8b}$. So firm 1 cheats.
- The cartel point lies off both best-response lines: each firm wants to move away from it.



# The payoff table

Each firm chooses its cartel share, $\frac{a-c}{4b}$, or its Nash output, $\frac{a-c}{3b}$. Profits (firm 1, firm 2), in units of $\frac{(a-c)^2}{144\,b}$:

| | Firm 2: cartel share | Firm 2: Nash output |
|---|---|---|
| Firm 1: cartel share | (18, 18) | (15, 20) |
| Firm 1: Nash output | (20, 15) | (16, 16) |

- **How to read it:** the first number is firm 1's profit; the second is firm 2's.
- **For firm 1, compare within a column.** If firm 2 keeps its share: 20 beats 18. If firm 2 does not: 16 beats 15.
- So the larger output is better for firm 1 whatever firm 2 does. The same holds for firm 2.

**Definition: Dominant strategy**

A choice that is best whatever the other player does. Here: the larger output, for both firms.

# The prisoners' dilemma

- Each firm's dominant strategy is the larger output.
- So both produce $\frac{a-c}{3b}$: the Nash equilibrium. Each earns 16.
- Had both kept the agreement, each would earn 18.
- Each firm's own best choice leaves both worse off.

**Definition: Prisoners' dilemma**

A game in which each player's own best choice leaves both worse off than if they cooperated.

**Finding**

The cartel agreement serves the pair, but it is not each firm's best response. That is why collusion breaks down.

# Part 3 summary

- Without a cartel, each firm maximizes its own profit, taking its rival's output as given.
- Best response: $q_1 = \frac{a-c}{2b} - \frac{q_2}{2}$, and the same for firm 2.
- Nash equilibrium: each firm produces $\frac{a-c}{3b}$ and earns $\frac{(a-c)^2}{9b}$.
- Against a partner who keeps the cartel share, a firm's best response is to produce more.
- Dominant strategy and prisoners' dilemma: both produce more, and both earn less.

# Check: Part 3

Cartels are difficult to maintain because

A the monopoly output is very difficult to determine.

B the number of firms is always large.

C costs to the firms in a cartel are continually rising.

D each firm has an incentive to deviate from its agreed output level.

# Check: Part 3 – answer

Cartels are difficult to maintain because

A the monopoly output is very difficult to determine.

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**Finding**

Against a partner who keeps the cartel share, each firm's best response is a larger output.

Part 4

# Market Outcomes Compared

# Three outcomes from the same demand and cost

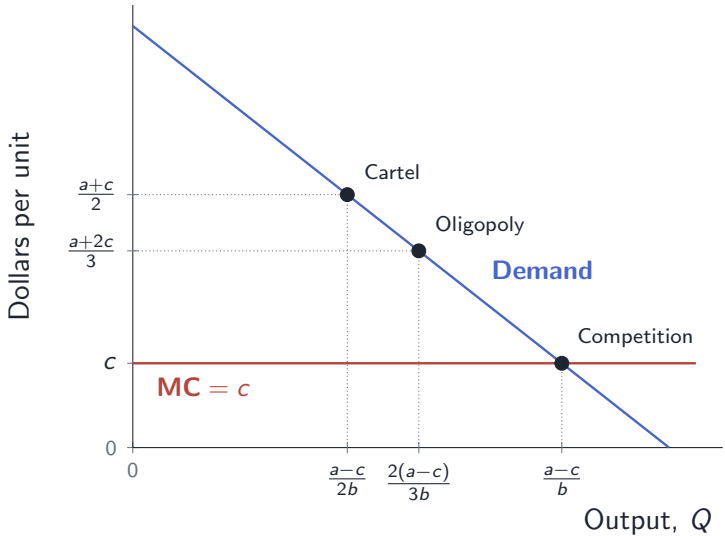
| | Total output | Price | Profit per firm | $P$ and MC |
|---|---|---|---|---|
| Cartel (monopoly) | $\frac{a-c}{2b}$ | $\frac{a+c}{2}$ | $\frac{(a-c)^2}{8b}$ | $P > \text{MC}$ |
| Oligopoly, no collusion | $\frac{2(a-c)}{3b}$ | $\frac{a+2c}{3}$ | $\frac{(a-c)^2}{9b}$ | $P > \text{MC}$ |
| Perfect competition | $\frac{a-c}{b}$ | $c$ | 0 | $P = \text{MC}$ |

Competitive output: set $P = c$, so $a - bQ = c$ and $Q = \frac{a-c}{b}$.

**Finding**

Output: $\frac{1}{2} < \frac{2}{3} < 1$ times $\frac{a-c}{b}$. Oligopoly lies between monopoly and perfect competition in output, price and profit.

# The three outcomes on the graph



# All four market structures in the long run

| | Perfect competition | Monopolistic competition | Oligopoly | Monopoly |
|---|---|---|---|---|
| Price and MC | $P = \text{MC}$ | $P > \text{MC}$ | $P > \text{MC}$ | $P > \text{MC}$ |
| Long-run profit | Zero | Zero | Can be positive | Can be positive |
| Why | Free entry | Free entry | Entry is difficult | Entry is blocked |

**Finding**

Only perfect competition sets $P = \text{MC}$. In the other three, some buyers who value a unit above its cost do not get it.

# Collusion and the law

- Collusion raises price and cuts output, so buyers lose.
- **Sherman Antitrust Act (1890):** agreements among competitors to fix prices or restrict output are illegal.
- **Clayton Act (1914):** individuals harmed by illegal agreements can sue to recover damages.
- **Example:** in 1996 Archer Daniels Midland agreed to plead guilty to fixing the prices of lysine and citric acid, two food and feed additives, and paid a $100 million fine.

Source: U.S. Department of Justice, press release, 15 October 1996.

# Part 4 summary

- Cartel: output $\frac{a-c}{2b}$, the lowest; price $\frac{a+c}{2}$, the highest.
- Oligopoly without collusion: output $\frac{2(a-c)}{3b}$; price $\frac{a+2c}{3}$.
- Perfect competition: output $\frac{a-c}{b}$; price $c$.
- Oligopoly lies between monopoly and perfect competition.
- Price-fixing is illegal under the Sherman Act.

# Check: Part 4

The equilibrium quantity in markets characterized by oligopoly is

A higher than in monopoly markets and higher than in perfectly competitive markets.

B higher than in monopoly markets and lower than in perfectly competitive markets.

C lower than in monopoly markets and higher than in perfectly competitive markets.

D lower than in monopoly markets and lower than in perfectly competitive markets.

# Check: Part 4 – answer

The equilibrium quantity in markets characterized by oligopoly is

A higher than in monopoly markets and higher than in perfectly competitive markets.

**B higher than in monopoly markets and lower than in perfectly competitive markets.**

C lower than in monopoly markets and higher than in perfectly competitive markets.

D lower than in monopoly markets and lower than in perfectly competitive markets.

**Finding**

$\frac{a-c}{2b} < \frac{2(a-c)}{3b} < \frac{a-c}{b}$.

# Which rule answers which question

| Question | Rule |
|---|---|
| What defines an oligopoly? | A few sellers, similar products, difficult entry, interdependence. |
| What does a cartel produce? | The monopoly output: MR = MC, $Q_C = \frac{a-c}{2b}$. |
| What is a best response? | The profit-maximizing output, given the rival's output. |
| What is a Nash equilibrium? | Each firm's output is its best response to the other's. |
| Why does a cartel break down? | Each firm's best response to the cartel share is a larger output. |
| Where does oligopoly output lie? | Between monopoly and perfect competition. |
| Is collusion legal? | No: price-fixing is illegal under the Sherman Act. |